CyberRiskGuardian in the Classroom

Instructor Guide — Education Package 2.1.2

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1 About this guide

This guide helps instructors, teaching assistants and program coordinators bring CyberRiskGuardian (CRG) into a course on cybersecurity risk management, governance, IT audit or business technology management. It explains what the method teaches, which version of the tool to use, how to fit it into a single workshop or a full term, how to run the labs, and how to assess the work fairly when students are allowed to use generative AI.

The package was built from the material that ships with CyberRiskGuardian (the method specifications, the governance textbook, the MediBec and CanaRetail worked examples and the desktop application's teaching features) and from the author's own undergraduate course on information systems risks, security and audit, where CRG has been used for the team risk-assessment assignment.

1.1 What is in the package

Component	Audience	Edition
Instructor Guide (this document)	Instructors, TAs, coordinators	Community (free)
Student Lab Workbook — thirteen labs on the MediBec case and a self-study practice case (CanaRetail), with self-check answers	Students	Community (free)
Assessment Kit — assignment briefs, 0–10 rubrics, self-checks, peer-review form, AI acknowledgement	Students and instructors	Community (free)
Case Library — five fictional business cases with a guide to choosing one	Students and instructors	Community (free)
Introductory slide deck — the CRG method in one class, with screenshots of the desktop application 1.9.1 (the 1.9.2 screens differ only in wording)	Instructors	Community (free)
Excel model v2-6 — the blank CyberRiskGuardian Excel model (`templates/`)	Students and instructors	Community (free)
Chapter slide decks — 21 decks for the textbook, one per chapter (GitHub release assets)	Instructors	Community (free)
Educator Edition — lab answer keys, graded quiz bank, rubric calibration and feedback bank, case teaching notes, the final-exam case and its blueprint, and the textbook's Instructor Resource Guide	Instructors only	Educational edition (Educational Licence)

Everything except the Educator Edition is published with the Community Edition under CC BY-NC 4.0, in English and French. The Educator Edition is kept out of the public repository so that answer keys and the exam case stay useful from one term to the next.

1.2 Who may use it, and when a licence is needed

- **Students** may use CyberRiskGuardian and this package free of charge for their own coursework, including work handed in for a grade.
- **Instructors** may read, adapt and reuse the free materials for non-commercial teaching, with attribution.

- **Colleges, CEGEPs and universities, as institutions,** need the CyberRiskGuardian **Educational Licence** to deliver a course built on CRG. It is offered at a significantly reduced price and includes installation, customization and the Educator Edition.
- **Non-profit organizations** may use CRG free of charge to assess their own risk.

Licence enquiries: **sales@cyberriskguardian.ca** · product site: **cyberriskguardian.ca**. The CC BY-NC 4.0 legal code shipped in the repository governs where this summary and the licence differ. This is not legal advice.

2 CyberRiskGuardian in one page

CyberRiskGuardian is a scenario-driven method for assessing cybersecurity risk, supported by an AI assistant (the Claude plugin and project), a desktop application and an Excel model. It is a **decision-support tool**: it structures the analysis and speeds it up, while the analyst stays accountable for every estimate and management stays accountable for every risk decision.

The workflow. Organizational context → crown jewels → risk appetite → 25–40 candidate scenarios (20 in the labs) → the 10 most material → six parameters and CVSS for each → estimated and tolerated risk → treatment portfolio → residual risk against tolerance → recommendations, budget and KRIs. The textbook (ch06) numbers these as the 19 CRG steps, followed by the management report and an independent quality review.

A scenario is a causal chain, never a threat name. *Threat source or event → vulnerability or weakness → affected asset or process → cybersecurity event → organizational consequence.* "Ransomware" is a threat; "a ransomware affiliate uses stolen VPN credentials and the absence of MFA on remote access to encrypt the clinical scheduling servers, stopping appointments at all sites for several days" is a scenario.

Six parameters, each on a 0–1 scale.

Symbol	Parameter	Question it answers
Pb(A)	Probability that the threat is present	Will this threat source act against us in the next 12 months?
Pb(ψ,A)	Probability of exploitation	If it acts, will it get through our weakness?
δe(ψ,A)	Expected damage	How much harm, reasonably expected?
δm(ψ,A)	Maximum damage	How much harm, plausibly at worst?
θ(ψ,A)	Organizational resilience	How well can we prevent, detect, contain, respond, recover and keep operating?
μ(E)	Organizational criticality (expected utility)	How important is the affected service to the mission?

The formulas (Excel Guide v1.0c, section 11 — authoritative, never changed silently):

```
Estimated risk  Re = Pb(A) × Pb(ψ,A) × CVSS × ((δe + δm) / 2) × μ(E) ÷ θ × Factor
Tolerated risk  Rt = Pb(A) × Pb(ψ,A) × CVSS × Appetite × μ(E) ÷ θ × Factor
Mitigated risk     = Re × Probability reduction × Impact reduction
```

```
Residual risk      = Re − Mitigated risk
Ratio              = Residual ÷ Tolerated
```

A ratio below 0.90 is read as **below tolerance**, 0.90 to 1.10 as **approximately at tolerance**, above 1.10 as **above tolerance**. The factor (usually 1,000) scales the scores; it never changes a ratio.

Three markers on every deliverable keep the evidence honest:

Marker	Meaning
Analytical estimate — validation required	A value the analyst (or the AI) estimated
Information gap — organizational validation required	A fact nobody has supplied
Management decision required	A choice only management can make

These markers are the single most useful habit the method teaches. A student who uses them correctly has understood the difference between evidence, inference and judgement.

3 Learning outcomes

CRG supports learning outcomes that most risk, security and audit courses already have. The table maps them to the labs and assessments in this package. Adapt the wording to your program's competency framework.

#	By the end of the course, students will be able to...	Labs	Assessment
LO1	Describe an organization's context, critical processes and crown-jewel assets from case evidence, separating facts from assumptions	1	A1 (context), exam
LO2	Estimate and justify a cybersecurity risk appetite and explain how it differs from tolerance and from a risk score	2	A1, quiz
LO3	Write plausible cybersecurity risk scenarios as complete causal chains and select a material portfolio	3	A1 (S1), exam
LO4	Estimate the six CRG parameters with rationale, evidence and confidence, and score a representative vulnerability with CVSS v4.0	4, 5	A1
LO5	Calculate estimated, tolerated, mitigated and residual risk and interpret the result against tolerance	6	A1, quiz
LO6	Design a cost-effective, non-duplicated treatment portfolio tied to the causal chain	7	A1 recommendation
LO7	Test how conclusions change under different assumptions	8	A1, exam
LO8	Define key risk indicators with owners, sources and thresholds	9	A1, A2
LO9	Communicate risk to senior management in a decision-oriented memorandum	10	A1 memo, exam
LO10	Use generative AI responsibly: verify its output, label estimates, protect data and disclose its use	All	All (professionalism criterion)

#	By the end of the course, students will be able to...	Labs	Assessment
LO11	Translate risk results into a security policy and a risk-based audit plan	12	A2, A3
LO12	Read an organization's identity, access and configuration posture from population-level evidence, and turn a finding into evidence and a justified parameter proposal	13	A2

4 Choosing a tool track

The method is the same in every track; only the interface changes. Pick one track per course and say so in the outline, so that teams are graded on the same footing.

Track	What students use	Strengths	Watch for
A. Claude (plugin or project)	The free CyberRiskGuardian Core plugin in Claude Code or Claude Cowork, or a Claude project carrying the CRG instructions	Fast scenario drafting, interview-style context intake, built-in label checks, `/crg-quick-scan` and `/crg-assess` commands, verified calculator	Students must verify AI output; institutional AI policy and data-protection rules apply
B. Desktop application	CyberRiskGuardian Desktop (current release; 1.9.2 Beta at the time of writing), served locally in a browser	Everything on one screen; a **next-step card** on the Process page that always says what to do and where; three detail levels (Minimal, Standard, Detailed); help linked to the textbook chapter of each screen; classroom workspaces; engine self-check; *Start over* for teaching cases; MediBec and CanaRetail bundled in English and French; the **Virtual consultant** (answers from the textbook, the ontology and the workspace, on a local model by default); optional **Identity & access management (IAM/PAM)** and **Configuration management** modules; simulation data for SIEM/SOAR exercises; EN/FR interface; no data leaves the computer unless the student sends it	Requires Python 3 to serve the folder; keep the same address and port so workspaces are found; one window at a time (a second window on the same address overwrites the first one's saves)
C. Excel model	The free CyberRiskGuardian Excel model (Analyse sheet, S1–S10, M1–M10), in `templates/`	Transparent formulas, familiar to business students, works offline	Manual calculation mode (press F9); validated inputs and a Checks column; CVSS blank and required; the appetite placeholder 0.5 must be replaced

A **hybrid** works well in practice: students draft context and scenarios with Claude (Track A), then quantify and treat in the desktop application or the workbook (Track B or C), where every number is recomputed by the verified engine.

4.1 Minimum set-up per track

Track A — Claude.

```
/plugin install cyberriskguardian --marketplace ITriskMgr/CyberRiskGuardian
/plugin install crg-governance-textbook --marketplace ITriskMgr/CyberRiskGuardian
pip install cvss "mcp>=1.2"
```

Restart Claude and approve the `crg-calculator` server when prompted. In claude.ai, the skills work too; calculations use the bundled `crg_calc.py` script.

Track B — Desktop application. Students follow the two first-time guides published with each release, in English and French: **First-time installation** (`FIRST-INSTALL.md` in the ZIP, and a PDF on the release page) and **First-time setup** (`FIRST-TIME-SETUP.md`, and a PDF). In short: unzip the release, then double-click `start.command` (macOS, Linux) or `start.bat` (Windows). The application must be served over `http://localhost:8099`; double-clicking `index.html` does not work. On the Process page, an optional card, *Set up the application for your practice*, offers **Answer the set-up questions**: one question about experience sets the **level of detail** (*Detailed* is the best choice for a first assessment) and the module questions choose the optional modules (students keep the defaults unless your course uses one, such as the two modules of Lab 13). Both can be changed later in Settings → Language & detail and Settings → Modules. Install it as an app from Chrome or Edge for offline use.

Track C — Excel. The blank model ships free with this package: `templates/CyberRiskGuardian_Excel_Model_Blank_v2-6.xlsx`. Post it on your learning platform. It follows the 8-step procedure of the textbook's chapter 6, printed on the Analyse sheet. Remind students that:

- it opens in manual calculation mode (Formulas → Calculate Now, F9);
- the appetite in E14 is a placeholder (0.5) that must be replaced with the case's approved or estimated value, strictly between 0 and 1;
- the factor in E15 is 1,000 and stays unchanged;
- the CVSS scores (G3:G12) start blank and are required, with the vector string next to each score (H3:H12); the Checks column (AC) flags a missing or malformed vector, an invalid input and a reduction error;
- on the M sheets, the reductions of the selected measures combine with the product rule $1 - \prod(1 - r_i)$, so a package can never exceed 1;
- the budget block (Y14:AC25) compares current spending plus the recommended portfolio with the appetite-consistent target and shows the gap and the implied appetite;
- columns AD and AE record the share of loss covered by a transfer and its approver, for information only.

4.2 The check every student does first

Whatever the track, the first thing a student does is confirm that the calculation engine agrees with the reference implementation:

- Desktop: the pinned footer of the side menu must read ✔ **Engine verified**; pointing at it, or Help → About & versions, shows the totals **84,305 / 41,023**.
- Claude or command line: from the plugin's `crg-quantify/scripts` folder, `python3 crg_calc.py ../../../evals/_shared/medibec30.json` must report **total estimated 84,305** and

residual 41,023 (in a clone of the repository, `examples/medibec/source/medibec30.json` is the same file).

- Excel: enter the textbook's chapter 6 worked example in S1 (CVSS 9.3, vector `CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:P/VC:H/VI:H/VA:H/SC:H/SI:H/SA:H`) and confirm estimated risk 3,554.46 and residual 1,937.18 after Calculate Now.

If it does not, nothing produced on that installation should be trusted. Make this check part of Lab 0 and ask for a screenshot.

5 Using the textbook

The package pairs with the textbook *Introduction to Cybersecurity Governance for Business Technology Management* (Léger, 3rd ed., v2.51b — the corrected v2.51 — October 2026), which is free as the `crg-governance-textbook` plugin (version 2.1.1) and whose appendix "Using CyberRiskGuardian with this book" maps all 21 chapters to CRG activities. The book exists in English and in French (*Introduction à la gouvernance de la cybersécurité*); the French documents of this package cite the French edition. In the desktop application, every screen's help opens with a **Read more: Chapter n — section** link to the passage of the textbook that explains it. The labs draw on these chapters:

Labs and activities	Textbook chapters
Lab 0 (engine check) and Lab 1 (context, crown jewels)	Chapter 6 (workflow, steps 1–3); Chapter 5 (context first)
Lab 2 (risk appetite)	Chapter 4 (appetite and tolerance, the tolerance bands, the budget benchmark)
Lab 3 (candidates, screening, selection)	Chapter 5 (25–40 candidates, five generation strategies, twelve screening criteria, coverage)
Labs 4–8 (parameters, CVSS v4.0, risk indices, treatment, sensitivity)	Chapter 6 (the risk index, CVSS by equivalency, the workbook procedure, the MediBec worked example)
Lab 9 (KRIs)	Chapter 6 (KRIs for monitoring); Chapter 14 (metrics and baselines)
Lab 10 (management memo, budget)	Chapter 6 (management report, budget calibration); Chapter 4; Chapter 17 (investment governance, insurance as transfer)
Lab 12 (from risk to policy and audit)	Chapter 9 (control frameworks), Chapter 10 (IAM), Chapter 3 (compliance and assurance)
Lab 13 (identity, access and configuration posture)	Chapter 10 (IAM and PAM, auditability and evidence); Chapter 14 (baselines, drift detection and remediation)
Practice case: CanaRetail (self-study)	Chapter 6 (the risk index, the workbook procedure); Chapter 4 (appetite-consistent budget)

Each chapter ends with twenty multiple-choice questions; the answer letters are in the book's appendix. The **Instructor Resource Guide** (rationales for every answer and marking guidance for the essay questions) and the 21 chapter slide decks accompany the book: the slide decks are free release assets of the repository (see `education/README.md`), and the Instructor Resource Guide is part of the Educator Edition.

6 Integration models

Choose the model that fits the time you have. Each model lists the sessions, the labs and a suggested weighting. The labs are described in the Student Lab Workbook.

6.1 Model 1 — One workshop (3 hours)

For an existing course that wants a single hands-on introduction.

Time	Activity
0:00–0:30	Introductory deck (slides 1–9): why scenarios, the six parameters, the markers
0:30–0:45	Lab 0: install or open the tool, verify the engine
0:45–1:30	Lab 3 (shortened): teams write three causal-chain scenarios for MediBec and critique each other's
1:30–1:45	Break
1:45–2:30	Labs 4 and 6: quantify one scenario and calculate its risk; compare with the worked example
2:30–3:00	Debrief: why the tolerance verdict depends only on damage, treatment and appetite; what management would decide

Assessment: a one-page reflection or the Lab 6 output (pass/fail).

6.2 Model 2 — Three-week module

For a course that dedicates a unit to risk assessment.

Week	Class	Labs	Deliverable
1	Context, appetite and scenarios	0, 1, 2, 3	Team scenario set (20 candidates in the lab version, 25–40 in a full assessment → 10)
2	Quantification, CVSS and interpretation	4, 5, 6, 8	Quantified portfolio with rationale
3	Treatment, KRIs and the management memo	7, 9, 10	One-page memo and two-page recommendation

Suggested weight: 20–30% of the course grade, using the A1 rubric of the Assessment Kit.

6.3 Model 3 — Full term (12 weeks)

This is the model used in the author's undergraduate course. The risk assessment (A1) feeds the security policy (A2) and the audit plan (A3), so students follow one organization from risk to governance to assurance.

Week	Course topic	CRG activity	Due
1	Governance, compliance, the security professional	Course overview; introductory deck; Lab 0	—

Week	Course topic	CRG activity	Due
2	Endpoint and network security	Lab 1: context and crown jewels (MediBec)	—
3	Risk management and privacy; performing a risk assessment	Labs 2, 3 and 4; teams receive their case	—
4	Threat landscape; security policies	Lab 5 (CVSS v4.0); teams draft their original scenario S1	—
5	Malicious code; social engineering; protecting information assets	Lab 6; team check-in on S1–S10	—
6	Security assessment and testing; cybersecurity audits	Labs 7 and 8	—
7	Cryptography and PKI	Lab 10 (memo clinic)	**A1 risk assessment**
8	Identity and access management; Zero Trust	Lab 9 (KRIs); A1 feedback	—
9	Resilience; monitoring and incident response	Lab 11 (optional, desktop): MediBec simulation pack	—
10	Cloud, wireless and mobile security	Lab 12, part 1: from risk to audit plan	—
11	Digital forensics; standards and certification	Lab 12, part 2: from risk to IAM policy; Lab 13: identity, access and configuration posture (evidence for A2)	**A3 audit plan** (individual)
12	Application security; review	Exam preparation; quiz bank (Educator Edition)	**A2 IAM policy**

Suggested weights, as in the author's undergraduate course: A1 20% (team), A2 10% (team), A3 10% (individual), final exam 40% (including an essay on a business case), quizzes and labs 20%.

6.4 Model 4 — Capstone or graduate seminar

Students work on a richer case (CNIC or ACRB) or, with the organization's written authorization and the data-protection precautions in section 9, on a real non-profit. Use the full scenario-selection procedure (Sections A–E: context, candidates, screening, final portfolio, quality review), the sensitivity analysis and the budget calibration. Require an oral defence of the portfolio before a panel acting as the board.

7 Before the term: a checklist

1. **Pick the track** (section 4) and test it on the machines students will use, including the lab computers.
2. **Pick the cases** (Case Library). Use MediBec for the labs, because its full worked example is public and students can check themselves against it. CanaRetail's solution is public too (in the desktop application),

so keep it for self-study practice. Assign Chez Pierre, ACRB or CNIC to each team for the graded assignment, and rotate cases from term to term.

3. **Prepare a classroom workspace** (desktop track): Organization → Workspaces → New workspace, *Start from* **Teaching example: MediBec — private healthcare** (English or French), type *Classroom business case*, tick *Educational case used for training purposes*. When it is ready to hand out, Organization → Reset data → *For the instructor* → **Set the current state as the starting point**, so that teams can start over.
4. **Decide which steps students redo** after *Start over* (by default, steps 2 and 3: context and crown jewels).
5. **Decide on the optional modules.** For Lab 13, students switch on *Identity & access management (IAM/PAM)* and *Configuration management* in Settings → Modules. The MediBec and CanaRetail workspaces already hold fictional sample registers for both modules; nothing in them changes a risk number.
6. **Load the simulation pack** if you plan Lab 11 (Settings → Simulation data → MediBec pack). The import report must show 40 alerts, 13 rules, 15 indicators, 4 cases and one refused self-approval (APR-MB-04).
7. **Write your AI rules into the outline** (section 9), including whether the Virtual consultant is allowed on each assignment, and attach the AI acknowledgement template from the Assessment Kit.
8. **Publish the Assessment Kit** with your due dates, page limits and weights.
9. **Check the institution's data-protection rules** for AI tools and decide which AI providers students may use. In the desktop application, each AI feature can be routed to a local model or switched off in Settings → AI — local & remote.

8 Running the labs

Each lab in the workbook states its objectives, timing, inputs, steps, deliverable and three self-check questions. The notes below are for the person running the session.

Lab	Time	What to watch for	A question that unlocks the room
0. Getting started	30 min	Students opening `index.html` directly (footer stuck on *engine loading…*); different ports losing workspaces; two windows open on the same address	"Why should you not trust any number until the footer says *Engine verified*?"
1. Context and crown jewels	60 min	Copying the case text instead of analysing it; inventing facts	"Which of these facts would change your risk ranking if it were wrong?"
2. Risk appetite	45 min	Confusing appetite (0–1 attitude) with a risk score; ignoring the value the case states	"MediBec states 0.40. Why might an analyst propose 0.30?"
3. Scenarios	90 min	Threat names ("phishing") instead of causal chains; ten variants of one technique	"Which step of your chain would a control interrupt?"
4. Quantification	75 m	Merging Pb(A) and Pb(ψ,A); δe greater than δm; θ set to zero; numbers without reasons	"Which two values are you least sure of, and what evidence would settle them?"

Lab	Time	What to watch for	A question that unlocks the room
	in		
5. CVSS v4.0	60 min	Scoring the whole scenario instead of one weakness; estimating a score by eye	“What vulnerability, exactly, did you score?”
6. Calculation	45 min	Reading the estimated risk as a dollar figure; ignoring tolerated risk	“Why does a rare scenario get the same verdict as a frequent one with equal damage?”
7. Treatment	90 min	“Control dumping”; counting a shared control twice; reductions summing above 1	“What does this control add that the organization does not already have?”
8. Sensitivity	45 min	Treating the central case as the truth	“Which decision changes between the optimistic and the pessimistic case?”
9. KRIs	45 min	Indicators without owners, sources or thresholds	“Who will measure this on the first of the month, and from which system?”
10. Memo	60 min	Restating tables; burying the decision	“What do you want the CFO to approve on Monday?”
11. Security operations (optional)	120 min	Treating an alert as an incident; using a source confidence as a probability	“Who confirmed this incident, and when?”
12. From risk to policy and audit	90 min	Policies full of technical values; audit plans that only check a policy exists; with the optional Writer step, handing in the generated draft as the answer	“How would an auditor prove this control works over a period?” · “Why did the Writer leave this decision as a placeholder instead of deciding?”
13. Identity, access and configuration posture	90 min	Reading a blank count as zero; reading a finding’s direction hint as a value; accepting a proposal in the workspace used for the other labs; naming people in register labels	“Which of these counts would you ask the IAM team to confirm first, and why?”

9 Teaching the hard concepts

Experience with the method shows that a handful of ideas cause most of the confusion. Teach them explicitly.

Appetite is not tolerance, and neither is a score. The appetite (0–1) expresses how much risk management is willing to pursue; the tolerated risk is the appetite applied to a specific scenario through the formula. The

textbook plugin answers “What is the difference between risk appetite and risk tolerance?” with a chapter citation — a good warm-up for Lab 2.

Threat presence and exploitation are two different probabilities. Pb(A) asks whether the threat source will act; Pb(ψ,A) asks whether it will succeed against this weakness given existing controls. Merging them double-counts.

CVSS scores a vulnerability, not a scenario. CVSS v4.0 Base measures the technical severity of one defined weakness. For phishing, insider, process, physical, third-party or governance scenarios, students score a *representative* weakness **by equivalency** (textbook ch06) and label it *limited applicability*. CVSS stays a mandatory multiplier, so a score is always needed; a malicious document delivered by e-mail is scored AV:L. A high CVSS never means the highest business risk.

The tolerance verdict depends only on damage, treatment and appetite. Residual ÷ Tolerated simplifies to ((δe + δm)/2) × (1 − P × Q) ÷ Appetite. Probabilities, CVSS, resilience and criticality cancel out: they drive the *size* of the risk (the ranking) but not the *verdict*. This is a striking result for students. Use it to show why the ratio must always be read together with estimated risk.

Before treatment, almost everything is above tolerance. The pre-treatment ratio is average damage ÷ appetite. With an appetite of 0.30, any scenario whose average damage exceeds 0.33 starts above tolerance. Students who expect a mixed picture before treatment are usually surprised.

Mitigation is multiplicative. An 80% probability reduction and an 80% impact reduction remove 64% of the risk, not 80%. Residual no greater than tolerated (ratio ≤ 1.0) requires P × Q ≥ 1 − appetite ÷ average damage; **below tolerance** (ratio < 0.90) requires P × Q > 1 − 0.90 × appetite ÷ average damage. A package that reduces only probability, or only impact, mitigates nothing under the workbook convention.

Shared controls are counted once. When one initiative serves several scenarios, its cost is split across them (an equal split is an acceptable default) and the portfolio total comes from the initiatives, not from the sum of scenario allocations.

Posture registers are evidence, not risk. The Identity & access management and Configuration management registers hold counts per population, never names. Their indicators (CRG-17 to CRG-25) are *measured* KRIs, not risk indices. A finding gives a direction (Pb(ψ,A) ↑ or θ ↓) and never a value: the analyst types the new value, the application shows the ratio before and after, and accepting it is a separate, journalled act. Use Lab 13 to show students how evidence enters an estimate without anyone “letting the tool decide”.

Two lenses may disagree. A simple normalized score (threat × exposure × impact × (1 − control maturity)) with escalation thresholds can place a treated risk within authority limits while the CRG ratio flags it above tolerance. Ask students to say which method a governance threshold is tied to.

10 Generative AI in the course

CyberRiskGuardian is itself an AI-assisted tool, so the course must take a clear position on AI. The position below is the one used in the author’s undergraduate course; adapt it to your institution’s policy.

Allowed on assignments, with an explicit acknowledgement (template in the Assessment Kit). Students are responsible for everything they submit, including anything an AI produced.

Not allowed during the exam. Students may prepare material with AI before the exam (for example risk scenarios submitted with the exam) if they disclose it.

What AI may do: draft scenarios, propose parameter values with rationales, critique estimates, suggest controls, check grammar, summarize sources.

What AI may not do: state facts about the case that the case does not contain, produce a final number that the student has not recomputed with the verified engine, invent standards, control identifiers, legal obligations, statistics or sources.

Verification is the skill being graded. AI tools invent control identifiers, CVSS vectors and laws with great confidence. Grade the verification, not the prose: a fabricated or untraceable source is both an evidence problem and an academic-integrity concern.

Data protection. Students do not paste confidential material into an external AI service unless authorized. Fictional cases are safe; real organizations are not, unless the organization has agreed in writing and the institution's rules allow it. In the desktop application, AI requests show the exact payload before anything is sent, can be anonymized, and can be routed to a local model.

The Virtual consultant: decide for each assignment. The desktop application includes a Virtual consultant (top-bar **Consultant** button, or Tools → Virtual consultant). It answers from passages retrieved on the student's computer — the textbook, the CRGO ontology, the workspace's documents and records — and keeps them apart from the model's general knowledge, which it marks *Analytical estimate — validation required*. It writes nothing into the assessment, and it removes any risk figure the engine did not compute. Because it is an AI tool, **state in each assignment brief whether it is allowed** (the A1–A3 briefs of the Assessment Kit have a line for this):

- **Allowed:** students declare it on the AI acknowledgement like any other tool, and attach the conversations they relied on (Tools → Virtual consultant → *History* → *Export as Markdown*). A conversation can also be cited in the evidence register (*Cite as evidence*), where it is classified as an assumption.
- **Not allowed:** students switch it off in Settings → AI — local & remote, setting the rows *Virtual consultant* and *Virtual consultant — parameter critique* to **Off**. Students work on their own computers, so this cannot be enforced technically. Treat it as an academic-integrity rule, and look at the workspace export: every AI call is recorded in the workspace's AI log, which Tools → Virtual consultant → *Decision log* shows.

By default the consultant runs on a **local model**, so no text leaves the computer unless the student routes it to a remote provider.

The markers are the bridge. Ask students to keep the three markers in every deliverable. A marker-free deliverable is a warning sign that AI output was taken as fact. The Claude plugin includes a label check (`check_labels.py`) that lists missing markers.

11 Assessment strategy

The Assessment Kit contains:

- the **0–10 scale** and how criterion scores combine into an assignment score;
- **briefs and rubrics** for the risk assessment (A1), the IAM security policy (A2) and the risk-based audit plan (A3), plus four shorter formats (lab portfolio, individual scenario paper, in-class quick-scan challenge, exam essay);
- **self-check lists**, a **peer-review form** for team assignments and the **AI acknowledgement**.

The Educator Edition adds the lab answer keys, a graded quiz bank with answers, rubric calibration examples, a feedback-comment bank, teaching notes for each case, the final-exam case with its blueprint, and the textbook's Instructor Resource Guide.

Protecting the integrity of the assessment.

- Use MediBec and CanaRetail for practice only. Their full answers are public: MediBec's in the published worked example, CanaRetail's in the desktop application (its Module D: the nine supplied scenarios and an original one, quantified and treated).
- Rotate the other cases (Chez Pierre, ACRB, CNIC) between terms and between sections for graded work. The Educator Edition holds a sixth case, reserved for the exam.
- Require the original scenario S1 to be distinct from every scenario supplied in the case and in the guides.
- Require the workbook or workspace export with the memo, and check that the numbers agree. Disagreement between files is the most common sign of unverified AI output.
- Ask for a short oral walk-through of one scenario's parameters from a randomly chosen team member.

12 Bilingual and inclusive delivery

The desktop application switches its whole interface between English and French (EN/FR in the top bar) while leaving case data in its original language. Its French terms follow the French edition of the textbook (*appétence pour le risque*, *joyaux de la couronne*, *registre des éléments probants*, *ratio de tolérance*), and MediBec and CanaRetail are bundled in both languages. All the free materials in this package exist in both languages. A French case remains French in an English interface; teams may write in either language if your institution allows it.

For accessibility, the labs can be done entirely with the keyboard in the desktop application and in Excel; the slide deck uses high-contrast text; every figure in the workbook is also given as text. Allow extra time for Lab 3 and Lab 10, which are writing-intensive.

13 Troubleshooting

Symptom	Cause	Fix
Footer stuck on *engine loading…*	The application was opened as a file	Start it with `start.command` or `start.bat`; open `http://localhost:8099`
A student's workspaces have disappeared	Different address, port or browser	Use the same address and port; import the last workspace export
Excel shows formulas but old numbers	Manual calculation mode	Formulas → Calculate Now (F9)
"θ required" or "CVSS required" in the Analyse sheet (columns M and N)	θ left blank or at zero on the S sheet, or the CVSS score not entered	Enter θ above zero (up to 1) and the CVSS v4.0 score with its vector
"Appetite required" in column N	The appetite in E14 is not strictly between 0 and 1	Enter the approved or estimated appetite
A scenario's ratio did not change after adding measures	Measures not applied, or only one dimension	Apply the package; add a measure for the other dimension

Symptom	Cause	Fix
	reduced	
“Review M…” verdict, or a warning under the TOTAL row of an M sheet	A coefficient (column K or L) is outside 0–1	Correct the coefficient; combined reductions use the product rule and never exceed 1
AI proposes a law the case does not name	AI invention	Remove it; cite only obligations the case names or that a primary source confirms
The simulation pack will not import	The workspace is not a MediBec copy	Import into the MediBec example or a copy of it
Footer reads ✕ *Engine check failed* (banner *Engine does not reproduce the MediBec baseline*)	The engine disagrees with the reference	Stop and restore a clean copy of the application
The Identity & access management or Configuration management screen says the module is off	Optional modules are off by default	Settings → Modules, group Governance
A second MediBec workspace appears after restoring a backup or importing a workspace file (1.9.1)	The bundled case was recognized by its identifier only	Fixed in Desktop 1.9.2 (restore and import recognize the bundled case). On 1.9.1, delete the extra copy in Organization → Workspaces

Fixed in Desktop 1.9.2. The issues below were met in the labs with Desktop 1.9.1 and are corrected in 1.9.2; the row tells you what students on 1.9.1 will still see and what to do.

Issue and labs (Desktop 1.9.1)	On 1.9.1	In 1.9.2
Lab 1 — Rules-only extraction wrote `[object Object]` into Name, Size, Region and Context, tagged INFERENCE and ticked (#18)	Untick the four values and type them from the case	Fixed: the extraction writes the text it found
Lab 3 — Promoting a candidate from the screening dropped its parameters, links and causal chain (#20)	Re-enter the six parameters, the techniques and the chain on the promoted scenario	Fixed: the promoted scenario keeps everything
Lab 5 — A stored CVSS score could disagree with its vector (#21)	Recompute the score from the vector in the editor before reading the risk	Fixed: the vector wins and a disagreeing score is flagged
Labs 10–11 — QC-02 failed every recommendation approved in the interface, and an empty approver was accepted (#22)	Ignore the QC-02 failure for interface approvals; always name the approver	Fixed: QC-02 reads interface approvals; no approval without a named person
All labs — Two windows open on the same address silently lost work (#2)	One window only (Lab 0)	Fixed: the second window is warned and read-only

Issue and labs (Desktop 1.9.1)	On 1.9.1	In 1.9.2
All labs — Restore and import created a duplicate MediBec workspace (#4, #5)	Delete the extra copy	Fixed
Lab 0 — Lab 0 [A] pointed `crg_calc.py` at a `medibec30.json` that is not in the scripts folder	Use the path given in Lab 0 (`../../../evals/_shared/medibec30.json`)	Documentation fixed in this package (2.1.2)

14 Support, feedback and updates

- Repository: **github.com/ITriskMgr/CyberRiskGuardian** (Community Edition, CC BY-NC 4.0)
- Product site and Educational Licence: **cyberriskguardian.ca · sales@cyberriskguardian.ca**
- Desktop application: the **Help** button of the top bar opens *Help on this screen*, *Step by step*, the *User guide* (English and French) and *About & versions*; the first-time installation and setup guides are on each release page

Report errors in this package through the repository's issue tracker, naming the document, the section and the language version. Suggested improvements from instructors who have taught with CRG are especially welcome.

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